

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	29.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Dysuria;Pelvic heaviness

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	39.0	%	20 - 40
Wbc	8840.0	cells/ μ L	4000 - 11000
Polymorphs	62.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.0	mg/dL	3.5 - 7.2
Blood Urea	23.0	mg/dL	7 - 20
Cue Pus Cells	10.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clarithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 400 mg infused over 30 minutes every 24 h)

Precautions: Obtain baseline creatine phosphokinase (CPK) and repeat weekly; hold or reduce dose if CPK rises >5× ULN. No dose adjustment needed for creatinine clearance >30 mL/min (patient Cr 0.7 mg/dL). Contraindicated in pregnancy and lactation. Monitor for signs of hypersensitivity or myopathy.

Rationale: Enterococcus faecium is predicted to be resistant to clarithromycin and only sensitive to daptomycin. Daptomycin is one of the few agents active against VRE and therefore is the appropriate choice for this recurrent urethritis caused by E. faecium.

Antibiotic Drug History

Clinical Summary

Infection Summary:

A 29-year-old female presents with recurrent urethritis, confirmed by pus swab culture. The causative organism is identified as *Enterococcus faecium*, a Gram-positive coccus with a higher risk of vancomycin resistance (VRE).

Resistance and Sensitivity Profile:

The organism is predicted to be resistant to clarithromycin and sensitive to daptomycin.

Recommended Antibiotic:

Daptomycin (6 mg/kg IV once daily, e.g., 400 mg infused over 30 minutes every 24 hours).

Rationale:

Daptomycin is the only predicted sensitive agent and is effective against VRE, making it the appropriate choice for

treating this recurrent urethritis caused by *E. faecium*.

Precautions and Considerations:

1. Obtain baseline creatine phosphokinase (CPK) and monitor weekly; hold or reduce the dose if CPK rises >5× ULN.
2. No dose adjustment needed for creatinine clearance >30 mL/min (patient's serum creatinine is 0.7 mg/dL).
3. Contraindicated in pregnancy and lactation.
4. Monitor for signs of hypersensitivity or myopathy.